

Exercise 4.10

Value iteration update for state values (Equation 4.10)

$$V_{k+1}(s) = \max_a \sum_{s',r} p(s',r|s,a) \times [r + \gamma \cdot V_k(s')]$$

To get analogous update for action values $\Rightarrow$ compute taking action a in state s followed by optimal actions

$$q_{k+1}(s, a) = \sum_{s',r} p(s',r|s,a) \times [r + \gamma \cdot \max_{a'} q_k(s', a')]$$

- $q_{k+1}(s, a)$: updated estimate of the action value
- $p(s', r|s, a)$: probability of transitioning to state s' with reward r after taking action a in state s
- r : immediate reward
- γ : discount factor
- $\max_{a'} q_k(s', a')$: optimal action value in the next state

This equation updates the value of the state-action pair (s,a) by considering the expected reward and the discounted value of the best possible action in the next state.